

```
R1(config)#do show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
R1(config)#int
```

```
R1(config)#interface gig
```

```
R1(config)#interface gigabitEthernet 0
```

```
R1(config)#interface gigabitEthernet 0
```

```
R1(config)#interface gigabitEthernet 0|
```

```
R1(config)#interface gigabitEthernet 0/
```

```
R1(config)#interface gigabitEthernet 0/0
```

```
R1(config-if)#ip
```

```
R1(config-if)#ip a
```

```
R1(config-if)#ip add
```

```
R1(config-if)#ip address 15.255.255.254 255.0.0.0
                                     ^
```

```
% Invalid input detected at '^' marker.
```

```
R1(config-if)#ip address 15.255.255.254 255.0.0.0
```

```
R1(config-if)#des
```

```
R1(config-if)#description ##sw1##
```

```
R1(config-if)#no shu
```

```
R1(config-if)#no shutdown
```

```
R1(config-if)#
```

```
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

In this step, I configured IP address 15.255.255.254/8 on Router R1's GigabitEthernet0/0 interface and enabled the interface using the no shutdown command. I then verified that the interface transitioned from down to up status.

```

R1(config)#interface gigabitEthernet 0/2
R1(config-if)#ip ad
R1(config-if)#ip address 20
R1(config-if)#ip address 201.191.20.254 255.255.255.0
R1(config-if)#des
R1(config-if)#description ##sw3##
R1(config-if)#exit
R1(config)#interface gigabit
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip address 182.98.255.254 255.255.0.0
R1(config-if)#de
R1(config-if)#des
R1(config-if)#description ##sw2##
R1(config-if)#exit
R1(config)#do show
R1(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       15.255.255.254  YES manual    up              up
GigabitEthernet0/1       182.98.255.254  YES manual    administratively down down
GigabitEthernet0/2       201.191.20.254  YES manual    administratively down down
Vlan1                    unassigned      YES unset     administratively down down
R1(config)#interface gigabitEthernet 0/2

```

Here, I configured additional router interfaces by assigning IP addresses to G0/1 and G0/2 and added interface descriptions for easier identification. After configuration, I used show ip interface brief to verify the assigned addresses and interface states.

```
R1(config)#interface gigabitEthernet 0/2
R1(config-if)#no shutdown
```

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
```

```
R1(config)#ip int
R1(config)#inter
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#no shutdown
```

In this step, I enabled GigabitEthernet0/1 and GigabitEthernet0/2 using the no shutdown command. I then confirmed that all configured interfaces were operational and showing an up/up status.

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
```

```
R1(config)#ip inter
R1(config)#ip interf
R1(config)#do show ip
R1(config)#do show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	15.255.255.254	YES	manual	up	up
GigabitEthernet0/1	182.98.255.254	YES	manual	up	up
GigabitEthernet0/2	201.191.20.254	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
R1(config)#
```

```

1#show ip interface b
1#show ip interface brief
Interface              IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0     15.255.255.254 YES manual up                  up
GigabitEthernet0/1     182.98.255.254 YES manual up                  up
GigabitEthernet0/2     201.191.20.254 YES manual up                  up
lan1                   unassigned      YES unset  administratively down down
1#write
Building configuration...
[OK]
1#show st
1#show start
1#show startup-config
Showing 807 bytes

version 15.1
!
service timestamps log datetime msec
service timestamps debug datetime msec
service password-encryption

hostname R1

ip cef
!
ip v6 cef

--More-- |

```

Here, I saved the running configuration using the copy running-config startup-config command. This ensures the router retains its configuration after a reboot.

In this screenshot, I reviewed the router's running configuration to verify that all interface settings, IP addresses, and descriptions were correctly applied. This serves as a final validation of the configuration.

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       15.255.255.254  YES manual  up          up
GigabitEthernet0/1       182.98.255.254  YES manual  up          up
GigabitEthernet0/2       201.191.20.254  YES manual  up          up
Vlan1                    unassigned      YES unset   administratively down down
R1#sh rn
R1#sh run
R1#sh running-config
Building configuration...

Current configuration : 807 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
```

```
!  
interface GigabitEthernet0/0  
  description ##sw1##  
  ip address 15.255.255.254 255.0.0.0  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/1  
  description ##sw2##  
  ip address 182.98.255.254 255.255.0.0  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/2  
  description ##sw3##  
  ip address 201.191.20.254 255.255.255.0  
  duplex auto  
  speed auto  
!
```

This view shows the completed interface configuration on R1, including interface descriptions and IP addressing for all three GigabitEthernet interfaces. The configuration provides connectivity to multiple network segments.

Here, I configured the PC's network settings by assigning a static IPv4 address and subnet mask. This allows the PC to communicate with the router on the same network.

PC1

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Port Status ☒ On

Link Speed ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0040.0B02.5A32

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 15.0.0.1

Subnet Mask 255.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::240:BFF:FE02:5A32

In the final step, I tested connectivity using the Command Prompt and successfully pinged the router's IP address. The successful replies confirm that the router and PC are correctly configured and able to communicate.

The screenshot displays the Cisco Packet Tracer interface. On the left, the 'PC1' configuration window is open, showing the 'Desktop' tab with a 'Command Prompt' window. The command prompt shows the command 'C:\>ping 182.98.0.1' and the output 'Pinging 182.98.0.1 with 32 bytes of data:'. On the right, the 'Simulation Panel' is visible, showing a list of events. The table below represents the data from the 'Event List'.

Time(sec)	Last Device	At Device	Type
0.000	--	PC1	ICMP
0.000	--	PC1	ARP
0.001	PC1	SW1	ARP
0.002	SW1	R1	ARP
Visible 0.003	R1	SW1	ARP

At the bottom of the Simulation Panel, there is a 'Reset Simulation' button and a checked checkbox for 'Constant Delay'. The bottom status bar shows '3. Configure the appropriate IP addresses on R1's interfaces, and enable'.